Scaffolded frame variants — test

1. `\quadframefirst`

$$2x^2 + 25x + 33$$
$$\begin{array}{cc} 2x & \\ x & \end{array}$$

$$2x^2 - 25x + 33$$
$$\begin{array}{cc} 2x & \\ x & \end{array}$$

$$3x^2 + 26x + 35$$
$$\begin{array}{cc} 3x & \\ x & \end{array}$$

$$3x^2 - 26x + 35$$
$$\begin{array}{cc} 3x & \\ x & \end{array}$$

2. `\quadframeouter`

$$2x^2 + 25x + 33$$
$$\begin{array}{ccc} 2x & + & 3 \\ x & & \end{array}$$

$$2x^2 - 25x + 33$$
$$\begin{array}{ccc} 2x & - & 3 \\ x & & \end{array}$$

$$3x^2 + 26x + 35$$
$$\begin{array}{ccc} 3x & + & 5 \\ x & & \end{array}$$

$$3x^2 - 26x + 35$$
$$\begin{array}{ccc} 3x & - & 5 \\ x & & \end{array}$$

3. `\quadframeinner`

$$2x^2 + 25x + 33$$

$$\begin{array}{ccc} 2x & & \\ & \times & \\ x & + & 11 \end{array}$$

$$2x^2 - 25x + 33$$

$$\begin{array}{ccc} 2x & & \\ & \times & \\ x & - & 11 \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{ccc} 3x & & \\ & \times & \\ x & + & 7 \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{ccc} 3x & & \\ & \times & \\ x & - & 7 \end{array}$$

4. `\quadframelast`

$$2x^2 + 25x + 33$$

$$\begin{array}{cccc} 2x & & + & \boxed{} \\ & \times & & \\ x & & + & \boxed{} \end{array}$$

$$2x^2 - 25x + 33$$

$$\begin{array}{cccc} 2x & & - & \boxed{} \\ & \times & & \\ x & & - & \boxed{} \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{cccc} 3x & & + & \boxed{} \\ & \times & & \\ x & & + & \boxed{} \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{cccc} 3x & & - & \boxed{} \\ & \times & & \\ x & & - & \boxed{} \end{array}$$

5. `\quadframebare`

$$2x^2+25x+33$$
$$\begin{array}{ccc} 2x & + & 3 \\ x & + & 11 \end{array}$$

$$2x^2-25x+33$$
$$\begin{array}{ccc} 2x & - & 3 \\ x & - & 11 \end{array}$$

$$3x^2+26x+35$$
$$\begin{array}{ccc} 3x & + & 5 \\ x & + & 7 \end{array}$$

$$3x^2-26x+35$$
$$\begin{array}{ccc} 3x & - & 5 \\ x & - & 7 \end{array}$$

6. \`quadframe`

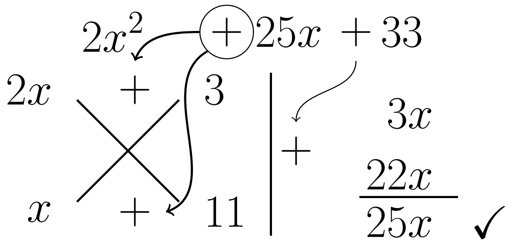
$$2x^2+25x+33$$
$$\begin{array}{ccc|c} 2x & + & 3 & \\ & & & + \\ x & + & 11 & \end{array} \quad \begin{array}{r} 3x \\ 22x \\ \hline 25x \end{array} \checkmark$$

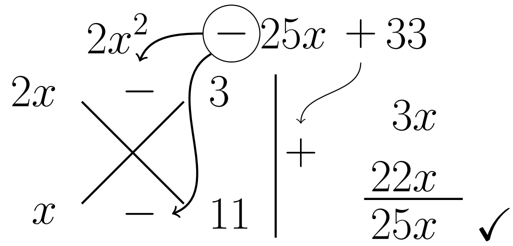
$$2x^2-25x+33$$
$$\begin{array}{ccc|c} 2x & - & 3 & \\ & & & + \\ x & - & 11 & \end{array} \quad \begin{array}{r} 3x \\ 22x \\ \hline 25x \end{array} \checkmark$$

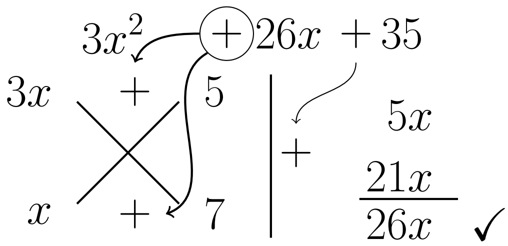
$$3x^2+26x+35$$
$$\begin{array}{ccc|c} 3x & + & 5 & \\ & & & + \\ x & + & 7 & \end{array} \quad \begin{array}{r} 5x \\ 21x \\ \hline 26x \end{array} \checkmark$$

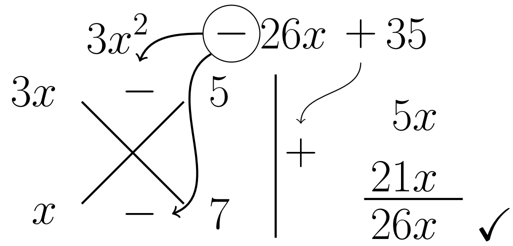
$$3x^2-26x+35$$
$$\begin{array}{ccc|c} 3x & - & 5 & \\ & & & + \\ x & - & 7 & \end{array} \quad \begin{array}{r} 5x \\ 21x \\ \hline 26x \end{array} \checkmark$$

7. \`quadframesign`









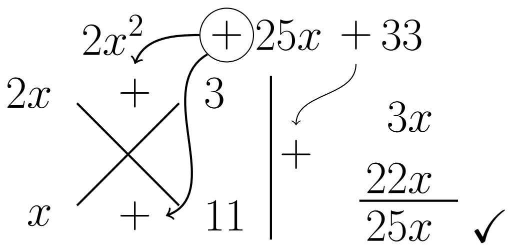
Factor each expression.

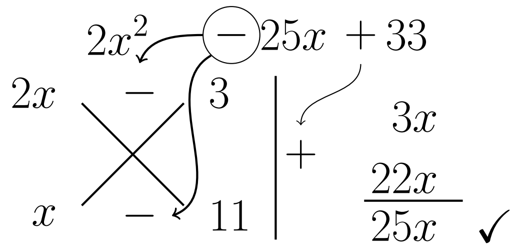
8.	x^2+3x+2	x^2+4x+3	x^2+6x+5	x^2+8x+7
9.	x^2+x-2	x^2+4x-5	x^2+6x-7	$x^2+10x-11$
10.	x^2+2x-3	$x^2+10x-11$	x^2+4x-5	x^2+6x-7
11.	x^2-x-2	x^2-2x-3	x^2-4x-5	x^2-6x-7
12.	x^2-3x+2	x^2-4x+3	x^2-6x+5	x^2-8x+7
13.	$x^2+13x+22$	$x^2+7x+10$	$x^2+33x+62$	$x^2+26x+69$
14.	$x^2+5x-66$	$x^2-3x-130$	$x^2-17x-110$	$x^2+8x-105$
15.	x^2-x-30	x^2-x-42	x^2-x-56	x^2-x-72

16.	$x^2 - 2x - 35$	$x^2 - 2x - 63$	$x^2 - 2x - 99$	$x^2 - 2x - 143$
17.	$x^2 - 3x - 40$	$x^2 - 3x - 70$	$x^2 - 3x - 108$	$x^2 - 3x - 154$
18.	$x^2 - 5x - 50$	$x^2 - 5x - 150$	$x^2 - 5x - 300$	$x^2 - 5x - 500$
19.	$2x^2 + 7x + 3$	$5x^2 + 36x + 7$	$3x^2 + 34x + 11$	$7x^2 + 15x + 2$
20.	$21x^2 + 10x + 1$	$10x^2 + 7x + 1$	$33x^2 + 14x + 1$	$65x^2 + 18x + 1$
21.	$2x^2 + 5x + 3$	$5x^2 + 12x + 7$	$3x^2 + 14x + 11$	$7x^2 + 9x + 2$
22.	$2x^2 + 5x - 3$	$5x^2 - 34x - 7$	$3x^2 + 32x - 11$	$7x^2 - 13x - 2$
23.	$2x^2 - x - 3$	$5x^2 + 2x - 7$	$3x^2 - 8x - 11$	$7x^2 - 5x - 2$
24.	$2x^2 + 13x + 15$	$3x^2 + 26x + 35$	$5x^2 + 62x + 77$	$7x^2 + 102x + 143$
25.	$2x^2 + 7x - 15$	$3x^2 - 16x - 35$	$5x^2 + 48x - 77$	$7x^2 - 80x - 143$
26.	$2x^2 + 21x + 27$	$3x^2 + 10x + 8$	$3x^2 + 40x + 125$	$8x^2 + 43x + 15$
27.	$2x^2 + 15x - 27$	$3x^2 + 2x - 8$	$3x^2 - 4x - 64$	$8x^2 + 43x + 15$

Factoring frame — worked examples

28.





$$2x^2 + 1x \quad - -55$$

$$\begin{array}{cccc|cl} 2x & & + & 11 & & \boxed{11x} \\ & \times & & & - & \\ x & & - & 5 & & \underline{10x} \\ & & & & & \quad x \ \checkmark \end{array}$$

$$3x^2 + 1x \quad - -10$$

$$\begin{array}{cccc|cl} 3x & & - & 5 & & 5x \\ & \times & & & - & \\ x & & + & 2 & & \underline{\boxed{6x}} \\ & & & & & \quad x \ \checkmark \end{array}$$

$$2x^2 + 21x + 55$$

$$\begin{array}{cccc|cl} 2x & & + & 11 & & 11x \\ & \times & & & + & \\ x & & + & 5 & & \underline{10x} \\ & & & & & 21x \ \checkmark \end{array}$$

$$3x^2 + 11x + 10$$

$$\begin{array}{cccc|cl} 3x & & + & 5 & & 5x \\ & \times & & & + & \\ x & & + & 2 & & \underline{6x} \\ & & & & & 11x \ \checkmark \end{array}$$

$$437x^2 + 1566x + 649$$

$$\begin{array}{cccc|cl} 23x & & + & 11 & & 209x \\ & \times & & & + & \\ 19x & & + & 59 & & \underline{1357x} \\ & & & & & 1566x \ \checkmark \end{array}$$

$$2x^2 + 25x + 33$$

$$\begin{array}{cccc} 2x & & + & 3 \\ & \times & & \\ x & & + & 11 \end{array}$$

$$3x^2 - 11x + 10$$

$$\begin{array}{ccc} 3x & - & 5 \\ & & \\ x & - & 2 \end{array}$$